6. PART D — REFLECTION QUESTIONS

i. Which wireless/WAN technology choice in your design was hardest to justify, and why?

The **VSAT/long-distance WAN connection between Windhoek and the Frankfurt global data center** was the most difficult technology choice to justify. VSAT can be more expensive and may have higher latency compared with terrestrial fiber or standard internet connections. However, Namibia's large geographical size and low population density make long-distance connectivity challenging, particularly in areas where terrestrial network infrastructure may not be readily available. We therefore selected VSAT as a simulated long-distance connection because it can provide reliable communication over very long distances without requiring a direct terrestrial fiber connection between Namibia and Europe. In our Cisco Packet Tracer simulation, the **serial WAN link** represents the VSAT connection between Windhoek and Frankfurt.

ii. How does Namibia's geography and population density shape the WAN/wireless choices you made for the Windhoek–branch link, compared to a similar link in a densely populated country?

Namibia's **large geographical size and relatively low population density** have a significant impact on how networks are designed. Connecting Windhoek with remote branches such as Walvis Bay can require substantial infrastructure, especially when using fiber, communication towers, or other terrestrial technologies. In more densely populated countries, network providers can serve more users within a smaller area, making fiber and mobile network infrastructure more cost-effective. For our design, we therefore selected different WAN technologies based on the requirements of each connection. The **Windhoek–Cape Town link** was designed as an **MPLS/cross-border WAN**, while the **Windhoek–Walvis Bay link** was represented as a **simulated site-to-site VPN**. This approach shows how factors such as distance, cost, infrastructure availability, and business requirements influence the choice of WAN technology.

iii. What cross-border challenges did you identify for the SADC link, and how might regional cooperation reduce these?

The main challenges associated with the **SADC link** include differences in telecommunications regulations, licensing requirements, network infrastructure, technical standards, and the cost of cross-border connectivity. Deploying fiber networks across national borders can also be complicated because it requires cooperation between different countries, regulators, and network operators. Regional cooperation could help address these challenges through **shared infrastructure, harmonised regulations, common technical standards, and agreements between telecommunications providers**. Countries could also work together to develop and maintain cross-border fiber routes. Such cooperation would improve regional connectivity, reduce infrastructure costs, and make it easier to maintain reliable communication networks across SADC countries.

iv. How did the global case study you researched change your understanding of what is technically possible versus what is currently deployed in Namibia?

The global case study, particularly the **DHL Global Supply Chain network**, demonstrated how large international organisations use a combination of networking technologies to build reliable, scalable, and efficient networks. Technologies such as **SD-WAN, cloud connectivity, hybrid WANs, edge computing, and centralised network management** provide greater flexibility and resilience than traditional single-WAN networks. This case study helped us understand that these advanced technologies could also be applied by organisations in Namibia. However, their implementation depends on factors such as infrastructure availability, cost, geographical conditions, business needs, and access to skilled networking professionals. Therefore, our design focuses on technologies that are practical for the Namibian environment while still incorporating concepts used in modern global enterprise networks.

v. What did using GitHub as a team teach you about version control and collaborative workflows in a real IT project?

Using **GitHub** helped us understand the importance of version control when working as a team on an IT project. Instead of storing separate versions of files on different computers, GitHub provides a central platform where project files can be uploaded, updated, and tracked. We learned how to make meaningful commits, organise files properly, assign tasks, and keep track of project changes. GitHub also showed us that effective teamwork requires good communication so that each group member knows their responsibilities and progress. These skills are valuable in real-world IT environments, where network engineers, software developers, and other technical professionals often work together on shared projects.

vi. If Kalahari Logistics doubled its number of branch offices next year, what would you change about your design to scale it?

If Kalahari Logistics doubles its branches, the network must prioritize scalability, automation, and centralized control. Implementing an SD-WAN solution will replace manual setups with simplified, multi-site management. A standardized IP addressing scheme will allow seamless expansion without IP conflicts. Furthermore, scaling dynamic OSPF routing and adding redundant WAN links will ensure high availability, while centralized firewalls, VPN management, and robust network monitoring will secure the expanded infrastructure.

vii. What was the most valuable lesson your group learned from this assignment, and how does it connect to a real-world career in networking?

The core takeaway from this project is that effective network design extends far beyond basic device connectivity; it demands a strategic balance of planning, security, scalability, budget, and environmental awareness. Utilizing Cisco Packet Tracer bridged the gap between theory and practice, allowing us to apply concepts like IP addressing, routing, WAN technologies, and wireless configurations in a simulated environment. Additionally, we discovered that troubleshooting is an indispensable skill, as physical layout correctness does not guarantee operational success. Ultimately, mastering this design, configuration, problem-solving, and security principles directly prepares us for the real-world responsibilities of a network engineer.